DISCUSSION

SESSION 4B

H. M. FINNISTON, *Session Chairman*

THE CHAIRMAN: In the design of fast reactors, all engineers consider their components to be of equal importance, but there is one which is more equal than all the others, and that is the fuel element. It is the fuel element with which we shall deal this afternoon in four papers—two by United Kingdom authors, one by American authors, and one from Euratom.

Presentation of Papers

J. F. W. BISHOP—*Paper* 4B/1

In the latter half of 1962, fuel element concepts were examined for a plutonium fuelled, prototype fast reactor with the requirements that they; should be capable of the high burnup needed for economic power production that they should provide modern steam conditions of 960–1050°F—the latter figure is 565°C; and that the reactor should have a good breeding ratio. A desirable condition was that only small amounts of fission products should be released into the coolant. The requirement therefore was for a design concept, a fuel material and a cladding material.

The three fuel materials considered were mixed oxide, mixed carbide, and metallic fuel.

With oxide, significant irradiation experience in a fast reactor had been acquired in the Russian reactor BR5, and relevant experience had been obtained by the American General Electric Company. We concluded that a sealed pin concept using stainless steel cladding was promising both from manufacturing and performance considerations.

With carbide, there was a lack of relevant irradiation experience, and considerable uncertainty about fuel cladding compatibility. Fuel manufacturing data were relatively sparse. A carbide concept was considered possible, but with substantial reservations attaching to it.

With metal fuel, no compatible fuel alloy and cladding combination was available with acceptable cladding corrosion in high temperature sodium.

So finally a sealed, mixed oxide fuelled pin concept was selected with a feasibility target of 5% burnup at a 600°C hot spot burnup, and with a requirement that 650°C hot spot and 10% maximum burnup should be indicated as attainable.

The selection of fuel form was dominated by the fact that condensible fission products would lead to about 1½% volume swelling per atom per cent burnup. This swelling would be inevitable, and would require local voidage of some 20% available to the fuel. Three fuel forms were considered; namely, solid pellets, annular pellets, and vibro-compacted fuel. We argued that vibro fuel and low density solid pellets would behave essentially in a similar fashion under irradiation; and that dense solid pellets would form a central hole on irradiation and then behave rather like annular pellets. So dense annular pellets and vibro fuel were chosen for development, with a smeared density of about 80%.

Cladding selection was based on the primary requirements for corrosion resistance and the potential to manufacture high quality tubing. The need to redirect fuel swelling, and to accept fast neutron damage of an ill defined nature, indicated that both strength and ductility considerations were important. Stainless steel and precipitation hardening nickel base alloys were selected for study. The subsequent detailed cladding development studies are reported in Paper 5B/3.

The chosen pin design criteria limit the total creep owing to 100% release of fission gas, to 0·1% integrated strain and specify that the combination of thermal and pressure stresses should not lead to continued incremental plastic deformation.

The choice of pin support within a wrapper was considered between grid support and helical finning of the pins. The important considerations were: the difficulty of producing finned or wire wrapped tubing, without incurring penalties on the metallurgical condition of the cladding or uncertainties of performance, for example stretching of loose wires; the accuracy to which pins could be positioned with the associated hot spot penalties; the problem of accommodating pin dimensional changes; and, obviously, the desirability of minimizing pressure drop. In the event, grids were selected, and we have under development both cellular honeycomb grids and strip eggbox grids as seen in Fig. D1b.

On the fuel manufacturing side, co-precipitated mixed oxide was chosen initially to give a consistent material, but, in bulk fuel handling, we may use physical mixing in due course: we think this will be essentially a plant decision. As far as the manufacture of stainless steel tubing, fuel pins and subassemblies is concerned, we are confident of being able to achieve a very high quality product. The briefness of my reference to this manufacturing work is not intended to reflect on the very substantial effort which has been expended, nor the crucial importance of this area of development.

The fuel element evaluation programme has comprised: firstly laboratory studies and associated environmental tests optimizing design detail and process and product specifications; secondly, environmental tests in sodium endorsing general mechanical and corrosion behaviour; and finally the very substantial irradiation programme, some results of which are discussed in Paper 4B/4. The achievements of that irradiation programme can be epitomized by the facts that we have taken a gridded subassembly to 5·5% maximum burnup (45,000 MWd/metric tons of fuel) without any structural deterioration and with not one of the seventy-seven fuel pins showing failures; also that similar pins have been taken successfully to about 8% maximum burnup.

In conclusion, therefore, my colleagues and I believe we can manufacture fuel elements consistently to the high quality needed for fast reactor service; we believe we shall achieve a low failure rate in fuel elements at 7·5% maximum burnup; we are directing our programmes to determining the overall combination of burnup, fuel and cladding material and fuel element dimensions to give the best overall economics in fast reactor fuel; we believe, finally, that mixed carbide fuel should be pursued vigorously both to formulate a successful design concept and to clarify its performance under irradiation, but we see no advantage at the present state of development in replacing oxide by carbide.

K. Kummerer—*Paper* 4B/2

Within our fast breeder project a major effort was started some years ago to develop suitable fuel elements for large fast power reactors. Mechanically mixed oxide was chosen as the fuel for the prototype reactors. The present paper describes shortly the logical structure of our work, the experimental activities, and some available results.

With regard to the objectives and the programme, the requirements are based upon the design studies for 1000 MW(e) sodium cooled and steam cooled reactors. In any case, a high burnup is envisaged. The maximum linear rod power lies between about 400 W/cm in the steam cooled case and 600 W/cm for sodium cooling.

In order to outline the programme we have established some working hypotheses, especially for the fuel parameters. Twelve such hypotheses can be found in the paper.

A lot of these hypotheses are highly conservative or founded on experimental results while others are theoretical and need practical proof.

D. A. Moss—*Paper* 4B/3

In the paper it was concluded that the ceramic fuels, primarily the plutonium–uranium compounds of oxides and carbides, had the near term

potential of meeting the requirements imposed by fast breeder reactor conditions.

Accordingly, the United States' programme is now concentrating on the two fuel types. Because of the many potential aspects associated with metal alloy fuels, however, development is continuing along this avenue also. The primary fuel on which we are concentrating is uranium–plutonium–zirconium alloy. The reason for this is the enhanced melting point temperature of around 1100–1200°C versus the old 800–900°C of the old fissium alloys.

February 1966

ORGANIZATIONS

AREA	ANL	PNL	GE	UNC	AI	W	BMI	NUMEC	ORNL	B&W	CARB'-DUM	GE-NMPO
CLAD	X	X	X		X	X			X			X
METAL	X											
OXIDE	X	X	X					X		X		
CARBIDE	X			X							X	
NITRIDE		X					X					
DR. FUEL		X										
OTHERS	X											
Na	X		X		X							

1 Private Funds

FIG. D1

ANL = Argonne National Laboratory
PNL = Pacific Northwest Laboratory
GE = General Electric Company
UNC = United Nuclear Corporation
AI = Atomics International
W = Westinghouse
BMI = Batelle Memorial Institute
NUMEC = Nuclear Materials and Equipment Corporation
ORNL = Oak Ridge National Laboratory
B & W = Babcock & Wilcox
CARB'DUM = Carborundum
GE–NMPO = General Electric NMPO

In addition, compatibility is enhanced to approximately 800°C with conventional stainless steels. To a lesser extent, mixed nitride fuel development has recently been initiated and is under way.

Probably the most important and yet most difficult task is that of developing a reliable cladding material, and a good portion of our effort is devoted to solving this problem.

Figure D1 presents a problem with regard to the organizations involved in our programme. I have listed these by areas of development—cladding,

metal alloys, oxide, carbide, nitride, and driver fuel. The driver fuel is implicit with the FFTF driver fuel.

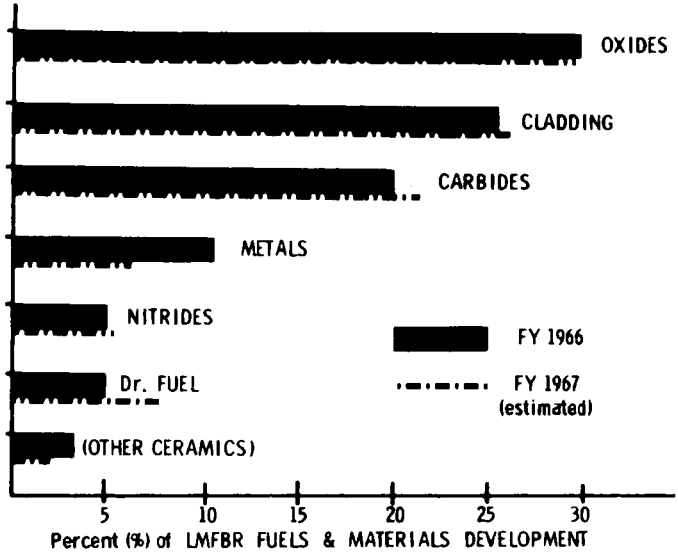

FIG. D2

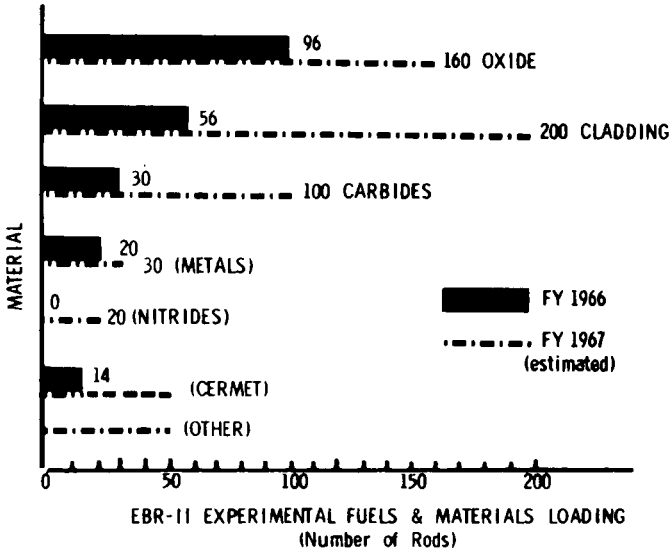

FIG. D3

Figure D2 indicates the relative level of efforts now applied to the various elements of U.S. fuels and materials development programme. The thick

black line represents this year's effort which ends June 30, 1966, while the thin dotted line represents the estimates for 1967. It is evident that our prime efforts are on oxides, almost 30% of our resources being applied here. Cladding, because of its importance, runs a close second. Driver fuel in direct relation to the FFTF—our prime emphasis is now on cermet but we are in process of looking at other driver fuel candidates such as oxides, metal, carbides. Any of them are candidates now.

The irradiations in the United States are carried out in thermal reactors and in fast reactors. The EBR-II is the prime irradiation facility now used and plans are under way to use the Fermi reactor. Figure D3 indicates the experimental irradiations broadly now in Fermi and those which are planned

AEC MATERIALS IRRADIATIONS
IN FERMI REACTOR

1. Space utilized	5 subassemblies
2. Temperatures	230°C (450°F) to 650°C (1200°F)
3. Exposures	10^{21} nvt (total) to $\sim 3\text{–}4 \times 10^{22}$ nvt
4. Number of specimens	~ 2400
5. Type of specimens	Tensile and tube
6. Type of materials	

(a) *Austenitic stainless steels* (60%)*
304, 316, 321, 347 and Incoloys
(b) *Nickel base alloys* (25%)*
Hastelloy X and Inconels
(c) *Refractory alloys* (10%)*
Vanadium alloys, columbium alloys and others
(d) Others (5%)*

* Per cent of AEC space utilized by material.

FIG. D4

for the next year. It should be noted that the irradiations listed are heavily supplemented by thermal reactor irradiations. What I have plotted are primarily the materials we currently have in EBR-II in addition to the number of rods in the reactor.

The emphasis again shows that, on our reactors, the problems are, primarily, in the area of cladding, and that the oxides have the best short term potential.

In addition we are currently planning to use Fermi as an irradiation facility (Fig. D4). The materials are primarily cladding, and of the eight subassemblies in the Fermi reactor we anticipate five will be materials. We are emphasizing stainless steels, and are looking seriously at the 304, 316, and the Incoloys, not discounting 321 and 347. A potential winner with the refractory alloys is one developed at the Argonne Laboratory which is a vanadium–titanium–chromium alloy.

Figure D5 shows our fuels programme in Fermi. Again we are concentrating on oxides and carbides. The oxide in category 1 and the carbide in

AEC FUEL IRRADIATIONS
IN FERMI REACTOR

Material	No. of rods	Parameters
1. Oxide	9	Vi-pac, 80 and 87% ρ*, O/M = 2·00, two exposures
2. Oxide	24	Pellets, annular pellets, Vi-pac, O/M = 1·97 & 200, cladding (304, 316, 347, Incoloy-800) 85 and 88% ρ*, three exposures
3. Oxide	8	Pellet, Vi-pac, mechanical designs to minimize fuel movement—transient tests later
4. Carbide	8	Vi-pac, 80 and 87% ρ*, 4·8 w/o C, two exposures
5. Carbide	12	Pellets, annular pellets, 80 and 85% ρ*, > 4·8 w/o C, porosity location
6. Metal	6	U-Pu-Zr, 70 and 80% ρ*, two exposures
7. Cermet	4	40 v/o PuO_2-SS, 70–95% ρ particles, transient test later
Total	72	

* Smeared density.

FIG. D5

U–20 W/O Pu NITRIDE IRRADIATIONS

Spec.	Fuel density	Fuel clad gap (mil)	kw/ft	Clad
1	95 (pellets)	< 1 (He)	25	Nb-1Zr (15 mil) 304-SS (15 mil)
2	85	< 1 (He)	↓	
3	95	10 (Na)		
4	95	20 (Na)		
5	85	10 (Na)		
6	95	10 (Na)		
7	95	20 (Na)		

40,000 and 80,000 MWD/Tu+Pu, 0,225 in. O.D. Fuel, 0·75 in. long
Also—accelerated BU capsule—0·15 in. O.D. Fuel

FIG. D6

category 4 are complementary in that they are both laboratory compacted, 80–87% theoretical density to investigate this parameter at two exposures. The oxide in category 2 is a pellet, primarily to investigate how best we can incorporate porosity in the fuel to avoid fuel swelling. The oxide in category 3 is primarily in support of our FFTF which will be looking at mechanical designs and other means to minimize fuel movement if it becomes a problem and they will be transient tested later.

The carbide in category 5 supplements the programme concerned with controlling porosity cracks. On metal, we are again contemplating different alloys.

We currently have four capsules going into a thermal reactor with uranium–plutonium nitride (Fig. D6). These four capsules have seven specimens and the temperature goes to about 900 up to 1200 and 1300°F.

We are looking at two types of material for cladding. One is 304 and it may be modified titanium. The other is a niobium–1% zirconium. I am not sure whether the 304 will fold up, but, in case it does, we will have the niobium–1% zirconium which we hope will run and give us the fuel irradiation information. We contemplate looking at them at 40,000 and 80,000 MWd/metric tons. To supplement this irradiation we have an accelerated burnup capsule which, because of its smaller diameter, will give us burnup approximately twice as fast, and we have a chance to look at the material on an interim basis.

PROPOSED EBR-II IRRADIATIONS (U,Pu) N

Rod No.	Bonding	Pellet dia., in.	Pellet density, %	Effective density, %	Cladding	Burnup MWd/t
1	Na	0·243	95	83	304	40,000
2	Na	0·243	95	85	Nb-1Zr	40,000
3	Na	0·243	95	83	304	80,000
4	Na	0·243	95	85	Nb-1Zr	80,000
5	He	0·255	85	83	304	80,000
6	He	0·255	90	88	Nb-1Zr	80,000

FIG. D7

Figure D7 represents six rods we are considering for irradiation in EBR-II as a supplement to the thermal irradiation programme you have seen. This again will look at 304 and niobium–1% zirconium so that we can correlate directly the thermal versus the fast flux irradiation. The prime emphasis on looking at the nitride now is with a sodium bond as we feel the compatibility of this material should be so much easier to deal with than with the carbide.

H. LAWTON—*Paper* 4B/4

This paper presents the results of irradiation experiments carried out in the Dounreay fast reactor to test possible fuels and fuel elements for the prototype fast reactor.

In addition to the observations made in the paper, I would like to divulge some new data which are the results of our examination of an assembly of seventy-seven mixed oxide pins, taken to an average burnup of 4·9% in the Dounreay fast reactor.

The pins were 53 cm long and 5·08 mm internal diameter, 5·84 mm external diameter—that is, the same diameter as most of the pins discussed in the

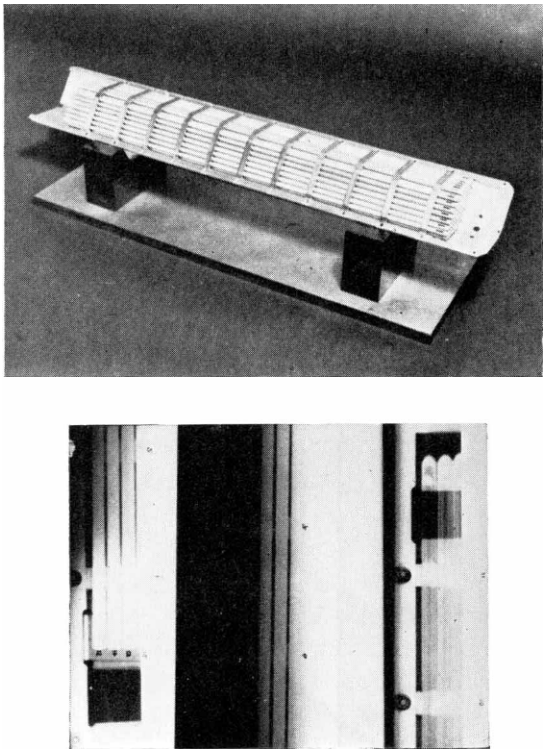

FIG. D8

paper. They were made of M316L stainless steel, and contained uranium–plutonium oxide fuel in either vibro-compacted or annular pelleted form. They were assembled together by grids providing a gap between pins of 1·25 mm. The grids were spaced 5 cm apart. The whole assembly was surrounded by two halves of a wrapper screwed together, and was irradiated at the core centre of the Dounreay fast reactor. Figure D8 shows the assembly with one-half of the wrapper removed. A maximum rating of 190 W/g was obtained, with a typical maximum can temperature of 550°C. The coolant flow rate through the assembly was 150 cm/sec. Thermocouples located beneath the assembly recorded maximum coolant temperatures.

After removal from the reactor, the assembly was allowed to cool in sodium for 80 days. On removal from the sodium, it was radiographed from many angles to look for pin failure or movement of components.

The assembly was then cleared of all traces of sodium–potassium coolant by blowing hot argon through it to which increasing amounts of steam were

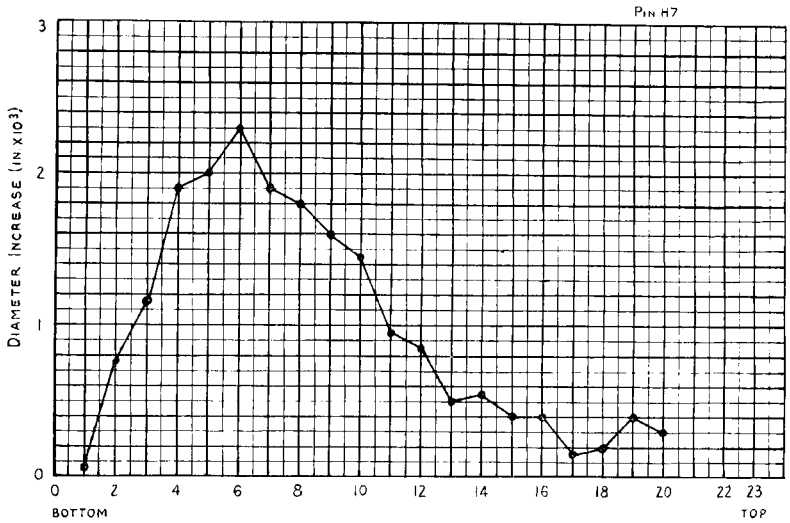

FIG. D9

added, finally finishing by rinsing in a water bath. Examination of the washings for plutonium indicated that pin failure had not taken place.

When the assembly was dismantled, it was seen that it had survived the irradiation very well. Grid deformation and corrosion was small, the wrapper was little distorted, and bowing of pins had been effectively restrained by the grids.

The pins were individually leak tested and all shown to be sound. Several of them were punctured and the released fission product gas measured. From pelleted fuel the release was in the range 39–45%, and for vibro-compacted fuel the range was 67–75%.

Each pin was radiographed and gamma scanned, but showed nothing unusual.

Measurement of pin diameter and length change provided valuable statistical evidence at this level of burnup. Pins extended in length by amounts ranging from 0 to 0·4%. Diameter increases were in the range 0 to 1·7%. These latter figures refer to the maximum diameter increases in each pin. This always occurred between the position of maximum flux and the position of maximum can temperature, generally some 16 cm from the bottom of the

pin. A typical plot of diameter increase along the pin is shown in Fig. D9. Clearly this sort of expansion can be ascribed to fuel swelling, since gas pressure swelling is greatest at the hottest point, i.e. at the bottom of the pin. It was not possible to draw any sharp distinction between the behaviour of pelleted and vibro-compacted pins.

More assemblies are included in our irradiation programme, and three can be accommodated in the centre of the reactor at once. One has just been removed after reaching an average burnup of 6·4% with some of the fuel at 6·9% burnup. One assembly now in the reactor contains some uranium–plutonium carbide pins.

Undoubtedly, this first assembly taken to 5% average burnup has provided the UKAEA with valuable evidence to support the case for building the prototype fast reactor, with much detailed information on seventy-seven pins, none of which had failed. Equally comprehensive data should be forthcoming at burnups in excess of 5% from assemblies now under irradiation.

Discussion

B. R. T. Frost—*Paper* 4B/4: I would like to add a few comments to Dr. Lawton's contribution on mixed uranium–plutonium carbide. In Dr. Bishop's paper and in other papers the role of carbide as a fast reactor fuel has been discussed briefly. In some of these papers it was indicated that the high fissile atom density of carbide might be attractive for breeding in fast reactors. At one time, we thought that the high thermal conductivity would be advantageous in maintaining low fuel temperatures and thus ensuring low fission gas release. However, as Dr. Lawton has shown for oxide, the predominant problem in fuel elements is not gas release but swelling, and thus retaining the gas within the fuel aggravates the swelling problem. I think this is illustrated by some results which we have obtained both at Harwell and at Dounreay on irradiation of 15% plutonium–uranium carbide.

All our experiments were on short pins, typically 0·7 cm in diameter, and they were intended to compare materials of different composition, density, and fabrication routes. Eleven of these pins were irradiated in the Dounreay fast reactor and they were clad in a Nimonic alloy which was chosen at a time when we were unfamiliar with the effects of irradiation on the mechanical properties of nickel base alloys. They achieved burnups between 4·5 and 6·5% with heavy atoms at centre temperatures around 1500°C.

Nine of these elements failed through fuel swelling combined with high thermal stresses. Two elements were powders of low density. These did not fail and only released 2% of their fission gases. A further twelve samples were irradiated in a thermal flux in one of the Harwell reactors. Their diameters were limited to 0·25 cm to minimize flux depressions and hence their centre temperature were less than 900°C.

The burnup range was 6·6–10·6% heavy atoms. At this low centre temperature, we observed low gas releases around 2% and no cladding failures. Two of the samples were of low density powders. These gave rather higher gas release of 6–18%, but these again did not fail.

It is interesting to compare these results with those obtained by Strasser and his colleagues in the United States in which they irradiated similar fuel to 13% burnup at a centre temperature of 1300°C. Under these conditions, gas releases around 30% were observed and all the stainless steel clad elements failed. The niobium clad elements gave swelling rates approaching 2% per 1% burnup.

The only evidence we have of larger scale irradiation of fuel pins comes from work at Dounreay where three mixed carbide pins 71 cm long with the same diameters as those Dr. Lawton quoted were irradiated in the DFR core. Two were filled with vibro-compacted powder obtained from arc melted stock with carbon equivalents in the range 4·78–5·01 wt.%. The third pin contained sintered annular pellets.

In all these three pins the mean density of the fuel stack was 80% theoretical. The surface temperature of the cold worked M316L cladding was in the range 560–655°C. One vibro pin reached 4·6% burnup without failure; the other vibro-compacted pin failed by cracking of the cladding at 7% burnup. The annular pellet pin reached 5·4% burnup without failure, so the burnup behaviour is rather similar to the oxide, although the level of failure is perhaps somewhat lower.

The important fact which emerges, however, is that the burnup limit for carbide is undoubtedly fuel swelling. Thus we can tentatively conclude that to achieve the necessary burnup in carbide fuel elements, we must either attempt to run the fuel hotter in order to release a high proportion of the fission gases and so minimize swelling or we must provide adequate space to accommodate the swelling without imposing excessive strains on the cladding.

A. DE STORDEUR (Euratom): I have a question about grid spacing concerning several results of the irradiations which Dr. Lawton has shown on the seventy-seven pin assembly. I understood that the grid spacing in that specimen was 5 cm, and the results show that they have not been distorting. Five centimetres is certainly too short a distance to be envisaged for a power reactor subassembly and I would like to know if distortion is expected for a larger spacing of the grids.

G. O. JACKSON—*Paper* 4B/1: The spacing of the grids in the Dounreay fast reactor (DFR) subassembly is determined by the rather large temperature rise per unit length of the pin that we have—about 230°C at inlet to about 500–600°C at the outlet—whereas in the PFR, the full length is longer, 36 in. against about 15 in. in the DFR and the temperature rise is less. It is 400° to about 600°C, so the spacing in the PFR would be about 4 in. or 10 cm

instead of 5 cm and we would not expect distortions to be worse under those conditions than in the DFR experiment.

A. DE STORDEUR (Euratom): Swelling seems to be a limiting factor on the burnup of ceramic pins and I think that everybody would be interested to hear some views on venting the fission gases to the coolant.

J. F. W. BISHOP—*Paper* 4B/1: As I outlined at the beginning of my talk, we were given the remit initially of considering a concept where release of fission products was on a very limited scale and I think it is common ground amongst all the people who are working on ceramic fuels—the Russians, the Americans, and the Euratom group—and therefore there clearly are design expedients of the lute kind which will delay the release of fission products and will minimize the activity problem. We ourselves are interested in the vented fuels because, whatever is the outcome of the swelling problem, we will certainly be pushed by the engineers to use the highest possible outlet sodium temperatures, and it is in this context that one would expect the benefit from the vented concept.

K. KUMMERER—*Paper* 4B/2: In the first development steps for the first series of prototype reactors, we should not aim for a vented fuel but there might be a vented fuel concept in a second approach; there are a lot of difficulties concerning radioactivity in the circuits, and as we are experiencing them now in the German association, we have made no major effort in the direction of fuel venting.

D. A. MOSS—*Paper* 4B/3: In the United States, we have a vented fuel under consideration but it is definitely not our prime effort. One of the biggest questions to be answered on this is how does this fission product, circulating through the system over the lifetime of a reactor plant, say 20 or 30 years, affect the overall maintenance and economic situation? I have a chart of some of our oxide capsule irradiations which are related directly or indirectly to vented fuel.

The B3A capsule was one that was irradiated approximately 2 or 3 years ago to about 14,000 MWd metric tons. The results indicated a hold-up time of about 5 days at least, which gave us an incentive to investigate further. We have other capsules, some just completed. The B4A has just come out and the results are not available, but as far as we know the capsule operated satisfactorily. The B4C has an intentional defect with a hole drilled to see what happens with sodium and there are other designs which we are pursuing.

We are not forced to make a decision immediately but are investigating various designs and capabilities of our venting techniques.

A. I. LEIPUNSKII (U.S.S.R.): Our experience in the small reactor BR5 is explained in more detail in Paper 2/3 but I will comment on our experience with reactive products of fission in sodium and in gas.

We made the experiment unintentionally because many of the pins were leaking, but we intentionally put them to high burnup. We get quite a lot of caesium in the sodium circuit; almost all the iodine was in the sodium, some of the chromium and barium, and, in the gas release, we had only the noble gases whereas in the fission products, a rather fast buildup of neon from sodium.

The life of neon in sodium is rather short. It goes through the gas blanket very fast. We have no figures about xenon and krypton. The figures on the amount of gas we get are only preliminary. We do not know exactly how many of our fuel elements are broken because they are small. There are figures in the paper but they are only as an indication. Certainly the amount of gas in the reactor did not interfere with its normal working, so we are looking forward to using leaking elements. But in spite of our good experience we have still designed the elements for BR350 with plenum as we do not want to use this kind of element in our first big power station.

I should like to ask a question now on fuel elements. I think it is generally believed that the most important part of a fuel element at the moment is the stainless steel tube. Current experience shows that the properties of steel, under irradiation and at high temperature, change a lot and not in the direction we want.

Our experience is only up to 10^{22} in integral flux, and in fast reactors we will need 10^{23} and even 10^{24} integral flux, and not only the fuel elements but the envelope will work under pressure all their life. It may be that the envelope will be in worse conditions than the fuel elements. Are completely new steels going to be developed for fast reactors for these very difficult conditions?

For our first reactor, we have no choice but to use ordinary commercial steels, but we have a feeling that they might not be right.

D. A. Moss: I think that without question our approach has been to use a conventional stainless steel. Our programme is divided into austenitic steels of approximately five alloys. If you figure out how long it has taken us to get as far as we have in the case of 304 and 316, it gives an idea of the sort of development problems that may be expected in changing materials. The incentive, therefore, in the United States is to use a commercial austenitic.

Another aspect of this is that an austenitic steel in America costs about 33 cents per linear foot. A refractory will cost anywhere from 6 to 20 dollars per linear foot. If you take a mixed oxide reactor such as GE has, there are 525,000 linear feet of tubing in it. This is an appreciable cost penalty to pay to move away from austenitics, and yet our uncertainty of its working is indicated by the fact that we have back-up programmes on the nickel alloys and on the refractory alloys and we are looking at five austenitics.

K. Kummerer: We feel also that we should use commercial austenitic steels. By commercial I do not mean that the type of steels should be fully

commercially available. One can develop a little bit in this region with additives to presently available stainless steels, although, as far as stainless steels are concerned, I should think that all the principal possible variations have already been tried in the last 50 years.

The alloy we favour now in our programmes, 16% Cr–13% Ni, is a specially designed alloy because it has some other additives than the normal material number, but as far as the other groups of alloy are concerned, the field is open. The development of nickel alloys is more or less just starting; this is also true of the vanadium alloys, but I would not recommend a vanadium base alloy and am hesitant about Inconel 625.

P. T. NETTLEY (UKAEA): The deterioration in the properties of stainless steel under fast reactor conditions is caused by the N alpha reactions resulting from reactions with thermal neutrons of boron and fast neutrons of the constituents of the steel. Therefore, there does not seem to be any real incentive to develop new steels. What we have got to do is to try and develop methods of heat treating or changing the metallurgical condition of the commercially available materials to make the deterioration caused by the fast neutrons as small as possible.

The United Kingdom therefore is devoting its effort to commercially available stainless steels and investigating methods of heat treating these materials to make the effects of irradiation as small as possible.

We are also doing work on Nimonic alloys because these have a higher strength than commercially available stainless steels, but a great deal more work needs to be done on the short term effect of irradiation, and on the creep strength and the ductility of all these materials, before a final choice is made.

W. R. MARTIN (Oak Ridge): I believe that most of us in the United States are either taking the opinion that we too can modify the commercially available steels in such a way as to negate the influence of the N alpha reactions. I think there is already suitable information in the literature to suggest that these approaches are rewarding. For example, the metallurgist can advise the users of his alloys to use a fine grain size material. This has been shown to improve the ductility of irradiated alloys both in short term properties and in creep properties.

In meetings to be held in our country in June, there will be data presented which will show other preirradiation heat treatments which vastly improve the ductility of the irradiated alloys, both in terms of spheroidizing the precipitates along the grain boundaries which retard the rate of grain cracks in the granular fracture, and in terms of reducing the helium content that eventually gets to the grain boundaries in fast reactor irradiations.

We are wholeheartedly supporting the stand that D. A. Moss (Paper 4B/3) presented earlier, that we would like to stick to commercial alloys. We know

in general how to fabricate them and their general corrosion properties, but we will make modifications to the alloys prior to irradiation to increase their mechanical properties to the maximum while in the reactor.

I want to ask Dr. Kummerer about his statement on the Inconel 625. The neutron exposure was about 3×10^{22} in which he observed no change in the mechanical properties. What was the preirradiation condition of the Inconel 625? And what mechanical properties were measured in which he observed no change and at what temperature were the measurements made?

K. KUMMERER: The properties which were measured during these experiments, preirradiation and postirradiation, were the tensile strength, at low temperatures and at temperatures up to 650°C.

W. R. MARTIN: With regard to the preirradiation conditions of the alloy, was it annealed at 625°C and with slight cold working in it?

K. KUMMERER: It was fully annealed. Different conditions were tested—fully annealed, and fully annealed with 20% cold worked before irradiation.

W. R. MARTIN: There were no changes like a reduction in ductility observed with a 600°C deformation temperature?

K. KUMMERER: There were some minor changes in the ductility but not important changes. If, for example, the tensile strength were in the region of 15 kg/mm^2, an important change would be 5 kg/mm^2—if the changes were in the region of ± 2 or 3 kg/mm^2. The reduction in ductile strength was about 10% or 15%.

A. G. FRAME (UKAEA): Would Mr. Moss outline his design philosophy of his metal fuel? Paper 4B/3 quotes burnups in the range up to about $4\frac{1}{2}$% at very low clad temperatures. Could he indicate (1) how he accommodates solid fission product swelling; (2) what he regards as limiting the clad temperature; and (3) what is his design target. He just said that the refractory alloys are twenty to fifty times more expensive than stainless steel: what does this mean in terms of his design target?

D. A. MOSS: We have currently in EBR-II, clad in 304, vanadium—20% titanium, and Hastelloy X, some twenty rods that are of uranium–plutonium–zirconium, or uranium–plutonium–titanium; I believe there are titanium pins in there. They are at about $1\frac{1}{2}$ at.% and going strong. We have now, leading this experiment, two pins clad in niobium–1% zirconium in a thermal reactor that are about at 6–7 at.% burnup. The temperatures on the cladding are ranging from around 650°C down. By down I mean that, as we burn up the fissile material, it will end up 5, 25, and 30°C down at the end of its life. They have been taken out and non-destructively examined by means of neutron radio-

graphy at 4·2 at.% and they looked good at the time. This constitutes some of our irradiation programme that gives us a little incentive in confidence limits.

With regard to the design philosophy, firstly, one would like to use a stainless steel, so what can one do, knowing the bad features of a fissium alloy?—in other words, there is a low temperature compatibility problem. One looks to see how to raise the melting point in the compatibility situation. We found that the zirconium and the titanium alloys do this. They raise it about 300° so, theoretically, one would expect to be able to put up the compatibility differential by about 300°. And so it turns out.

At 850°C we see melting. Metal fuels should be weak at these temperatures. To clad them in stainless steel is our primary objective. The design philosophy constitutes investigating these materials at about 10–14 kW/ft with the intention of pushing it up. Our design also constitutes a 70–80% fuel and the remaining portion void volume.

We shall try the metal fuels for another 2 or 3 years to see if we can develop an adequate concept and get meaningful irradiations and then have another hard look at them.

J. F. W. BISHOP: All countries have spoken about 100,000 MWd/metric ton and we do not mean 100,000 MWd/metric ton regardless of the design that will get there, so we have a balance between the materials and what the detailed fuel element design will achieve. We are very interested in the American work, the studies going on at the Argonne Laboratory, because it has been indicated that there may be a high temperature range in which there is some hope of getting a cladding material which does not corrode severely in high temperature and sodium which is compatible with the fuel material. What is obviously less clear, and I presume it is the major intention of the American programme to find out, is the burnup which this will go to, and in fact, Mr. Moss did not answer Mr. Frame's question—What sort of burnup target have you got to reach?

D. A. MOSS: In our system, the burnup question is going to depend primarily upon what the fuel cycle system is. What may be required for an oxide of 70,000–100,000 MWd/metric ton probably will not be required for a metal fuel. All I can say, therefore, is that basically we are striving to get fuel burnup data on a reliable basis that will get into the 50,000–70,000 MWd/metric ton range. This means it may be possible to accept a low temperature rapid recycling fuel temperature system which may give you the same as a high temperature burnup system. At 50,000–70,000 MWd/metric ton we will take another look at it, I do not think there is any clear-cut answer.

E. C. W. PERRYMAN: We have not heard anything about the movement of plutonium. This, I would have thought, would have been of real interest

to anyone talking about fast reactors. I have two illustrations which may be relevant to the movement of plutonium in uranium carbide elements.

A number of SAP clad uranium carbide elements were irradiated in an organic loop. They were filled with argon and the results I want to show refer only to hypostoichiometric fuel of a composition of 4·65 carbon. Before irradiation, the uranium in this fuel was uniformly distributed. They were arc-cast pellets and the uranium was mainly in the grain boundaries. After

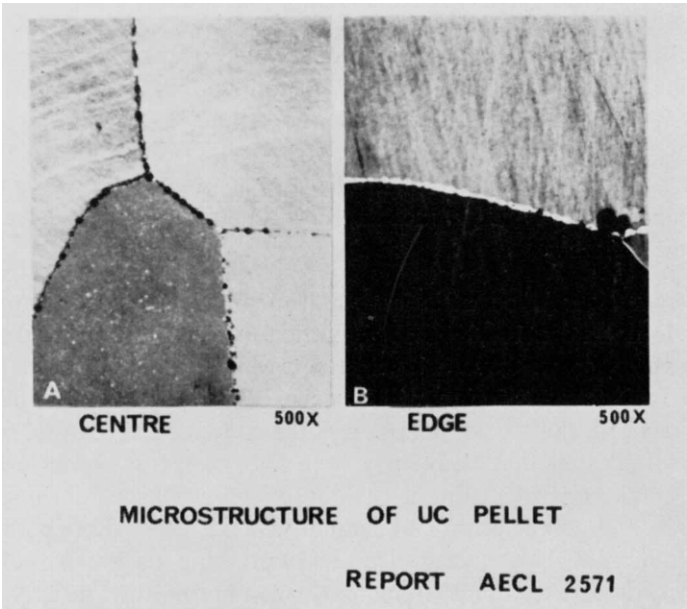

FIG. D10

irradiation there was movement of the uranium from the centre of the fuel to the periphery and also the pellets were dimpled. They had a sort of hemispherical seat. There was a void space in the fuel. We found that this uranium formed a deposit on the void spaces and it looked as though the grain boundaries were acting rather like a channel for the uranium to find its way to this void spacing.

Figure D10 shows, on the left, the structure in the centre of the element, and the black dots on the boundaries are holes. On the right is a uranium film. In Fig. D11, at the top, is the deposit on the top of the pellet, and on the right-hand side is the uranium threading up through the grain boundaries of the pellet.

The Americans say in Paper 4B/3 that they have not done any metallo-

graphy on their irradiated uranium carbide plutonium carbide fuel. Perhaps this will give them an opportunity to tell us what they have found.

Going back to stainless steel, I would like clarification on one point. In Paper 4B/3 the authors talk about the importance of the temperature of straining the material and they quote 500°C. In other words, above 500°C they

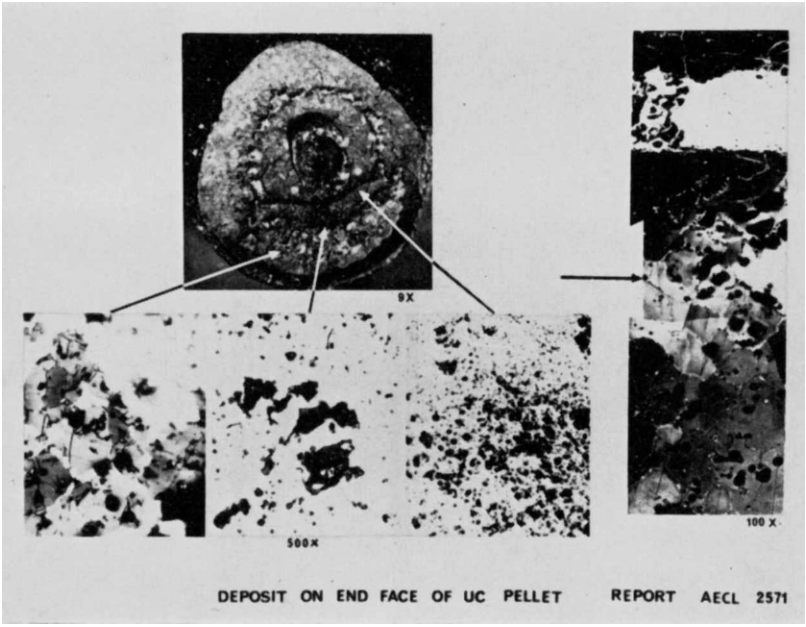

FIG. D11

say that you get this low ductility. The United Kingdom workers report 650°C. Which temperature is right?

W. R. MARTIN: The temperature at which you first observe the embrittlement due to N alpha conditions is surely a function of the metallurgical structure of the alloy itself and other deformation conditions, such as whether or not one is running a short time tensile test or one is running a stress rupture test. Also, it is a function of the dose level in which one is operating. With this in mind, all our information is up to neutron doses of about 3×10^{21}. In short time tensile testing, with strain rates of the order of 0·2% per minute, we first observed embrittlement of the stainless steels at about 600°C. This is with respect to our 18–8 unstabilized steels and I think that this is applicable to all our grades of austenitic steels in our country.

In creep, it appears that we do not get embrittlement at temperatures of about 550°C at stresses in the order that produces rupture in around

1000–5000 hours. There is some indication that this might be as low as 500°C. Herein lies our reason for selecting the temperature of about 500°C. I think this is also true in general for the nickel base alloys. We have observed the N alpha embrittlement there for these systems at 550°C and possibly as low as 500°C.

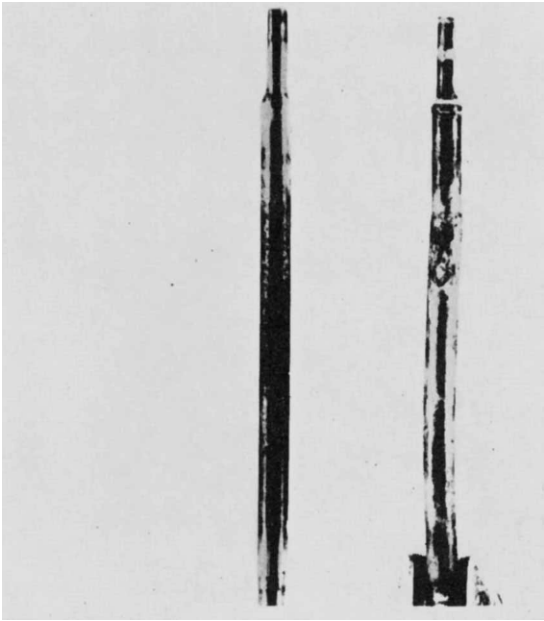

FIG. D12

P. T. NETTLEY: My answer to Mr. Perryman's question is that the temperature at which you see the effect depends upon the type of steel used, the dose that you give the steel, the flux spectrum that you put the steel in, and the temperature of irradiation. These four parameters are important. There are probably others. All we say is that for our steels, irradiated with the doses we quote for the Dounreay fast reactor in Paper 5B/3, we see the effect coming in at the 600–650°C level.

D. A. Moss: With regard to the particular problem of when the embrittlement takes place, it is interesting to note that in our EBR-II, where we are irradiating a metal fuel, one of the criteria for stopping irradiating is when the fuel swells out to the edge of the clad so there is no pressure on the cladding material. This is with 9 mil stainless steel. Somewhat in support of the 500°C we have seen at about 550°C about 0·5% elongation. At 500°C, we see about 1% but there is not a lot of difference between 1% and 0·5%. This is the temperature range in which we feel this mechanism is taking place.

Figure D 12 shows 15–20 wt.% plutonium carbide that was vibro-compacted

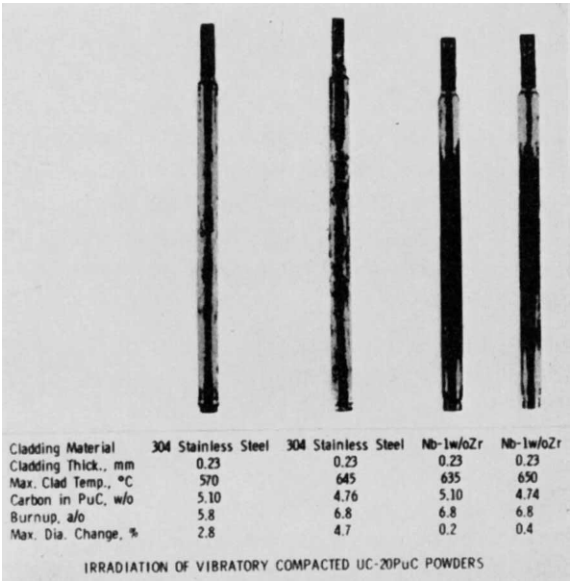

Fig. D13

into a uranium carbide matrix. The white material is the plutonium carbide; it is hypostoichiometric and one can see free plutonium; and the darker is the uranium carbide; it is closely stoichiometric and does not contain free uranium. The important thing to note here is that this fuel specimen was taken to around 5·2 or 5·8 at.% burnup, and all the fissioning was taking place in the plutonium carbide, so that this receives some forty-five times more fissioning than you normally see, and note the porosity and the plasticity of this material.

Figure D13 shows some of the material that Fig. D12 was taken from, at 5·8 at.% burnup. There was a series of six of these rods. Four are shown here and two were plutonium carbide. One of the plutonium carbide

ones failed, but one of the more important things to note is that this material did swell. It is 23 mm and about 9–10 mil stainless steel. It swelled 2·8% and 4·7% and you might expect this at a higher temperature, one being more hypostoichiometric than the other. Note the Nb–1% Zr claddings which operate at very high temperatures, but it was pretty good in restraining the fuel throughout.

B. R. T. FROST: One specimen we irradiated in thermal flux with 15% PuC–UC contained about 4% carbon, so the amount of free metal in the initial microstructure was considerable but evenly distributed throughout the structure. The centre temperature under irradiation was estimated to be 1400–1500°C, generally well above the melting point of the metal phase. After irradiation, almost the entire central region of the fuel was devoid of the metal phase which had migrated to the colder outside regions of the fuel, and obviously this would aggravate any compatibility problem between the fuel and the cladding. It merely reinforces our feeling that hypostoichiometric carbide is a bad thing to use anyway. We have no direct evidence from this experiment of gross migration of plutonium along the thermal gradient. We do not think that there is any serious migration, judged from microstructures, because we would have expected to see far more fission in the region to which the plutonium had migrated.

E. WAIT (UKAEA): We have been making estimates of the chemical state of irradiated carbide fuels at fairly high burnups, say around 10%. Our conclusions—and these are very preliminary conclusions—are that the situation is complicated but that when one starts with a monocarbide phase, stoichiometric uranium–plutonium carbide, at the end of the irradiation, the carbon, which has been released by the destruction of the plutonium, is more than sufficient to form carbides with those fission products which form carbides more stable than U–PuC; but that there is insufficient carbon available to form carbides with all the molybdenum and the strontium and barium that are present; so in the case of a monocarbide, there is very little likelihood, at least on a simple analysis, of releasing free metal.

In the case of a hyperstoichiometric carbide, again we have the rare earth carbides, zirconium–niobium carbides, formed. These are very stable but there is then a large excess of strontium and barium and molybdenum carbides which, while less stable than the monocarbide, are more stable than a sesquicarbide. Therefore, the effect of irradiation on a hyperstoichiometric material is to render it slightly less carburizing.

We do not know the relative stabilities of whatever phase is formed with the molybdenum and technetium and the strontium and barium dicarbide, but, in both cases, it seems likely that the carbon activity of the resulting fuel fission product mixture is controlled by one or other of these equilibria. In the case of the oxide, the analysis is a little more certain. The situation here

is that we have a group of fission product oxides, zirconium oxide, the rare earths, the strontium and barium, which are considerably more stable than the uranium oxide and UO_2+X regions of the phase diagram. These are likely to form oxides in the circumstances. On the other hand, we have a group of oxides like caesium and the known metals which are very unstable and do not really enter into our situation; so in the case of a uranium oxide fuel, the amount of oxygen which is released in fission is just about sufficient to form oxides with the stable oxide formers, provided the rare earth oxides enter the uranium dioxide phase in solid solution. The oxygen which remains probably goes to either oxidize the stainless steel container or to oxidize a little of the large excess of molybdenum which is still present. In this case, it is likely that the oxygen activity of the irradiated fuel is controlled by the molybdenum–metal dioxide equilibrium.

In the case of a fuel where most of the fission arises from plutonium, however, there is a difference because of the difference in the fission yield distribution of plutonium compared with uranium; considerably more oxygen is available. Probably most of the molybdenum will be converted at least to the dioxide and possibly to a higher oxide—trioxide—and, in fact, the oxygen activity of the resulting fuel is rather sensitive to the starting composition of the oxide. In the case of the uranium dioxide, the oxygen activity at the end of the irradiation is quite insensitive to the starting composition of the oxide.

S. LAWROSKY (Argonne National Laboratory): The information obtained from the irradiation results at various places, particularly including Dounreay, is reassuring as regards the low incidence of fuel failure. Have Dr. Lawton or Dr. Bishop any opinions as regards fuel failure propagation?

J. F. W. BISHOP: Dr. Lawton reported in Paper 4B/4 some results on pins which had been run on in the reactor beyond the point for which they were originally designed, and he mentioned that these failed in the reactor. These were pins V.021, V.024 and A.153. In the cases of V.021 and V.024, which were short, 7 in. pins, they were not in flowing sodium but static sodium inside a capsule. In the case of the failure of A.153, the failure was in a pellet pin at very high burnup and it is shown in the paper that the failure was at three places simultaneously. In that particular irradiation, the sodium flow was downward and it was not of high velocity because the vehicle was designed to reach the temperatures required which were higher than the outlet temperature from the reactor, and the situation with all of these pins which are mentioned—the one in very low velocity sodium, the one in downward flow, and the two static capsules—was that the mode of failure was surprisingly clear. You see in the paper the actual situation of the pin after failure.

G. O. JACKSON: The mode of failure is a point of some importance in the design of subassembly, and the worry here is that the gas from the plenum comes out rapidly through a small opening and blanks neighbouring pins, causing propagation of the fault to neighbouring pins, and further propagation from that pin to other pins, and then you very soon have real trouble in a subassembly. The encouraging thing that we have found about the failures we have had is that we do not, from the form of these failures, think that we will get this type of thing happening, nor do we feel, from the type of fault that has appeared, that we will get fuel coming out of the pins unless one progresses very much further with the irradiation after the failure has occurred.

J. HOCHEL (Germany): When making these rather high burnup experiments at Dounreay, have observations been made on the alteration in thermal conductivity dependent on the burnup? Or the dependence of thermal conductivity for mixed oxides on the plutonium content? Perhaps someone could comment on the alteration of the melting point depending on the content of plutonium and on the burnup respectively.

H. LAWTON—*Paper* 4B/45: We have very little to say on thermal conductivity because we cannot effectively make direct temperature measurements in our pins. This is because of our inability to get thermocouples out of the reactor. This means that I cannot comment on the question as to the effect of plutonium content on thermal conductivity nor indeed on the effect of fission products, etc., on the melting point.

D. A. MOSS: We have some data from the United States where we have irradiated, with a gas bulb temperature thermometer, two specimens to around 7000 MWd/metric tons. One specimen was of mixed oxide, 20 wt.% plutonia. The other was urania.

The data show that in the 800°C range, you are talking about 3/100 W per cm °C for the mixed oxide, about 0·35 for UO_2. The interesting feature is that the two lines blend together once you get up above about 1200°C. I think that possibly the plutonia does not have too much effect on the thermal conductivity of this. One of the more interesting aspects we have found that affects it is the stoichiometry. You can pick yourself up 200° in melting piont temperature and potentially a small fraction in thermal conductivity by using O to M ratio. By the same token, I think there are Westinghouse data in the United States where we have taken fuel at 40,000 MWd/metric ton range, and if I recall right, at 10,000 MWd/metric ton you can expect a decrease around 38°C. I believe you can expect a decrease in melting point temperature as the fission products build up in the fuel system. We are in process now of trying to figure out how best to take some of this high burnup fuel we have and make a melting point determination on it.

E. C. W. PERRYMAN: At the end of Dr. Lawton's paper (4B/4) he shows a graph which shows the diameter of the columnar grain growth zone in the centre of the fuel element increasing with burnup. I thought perhaps the interpretation of this was a change in thermal conductivity. If it is not this, what is the explanation?

K. Q. BAGLEY—*Paper* 4B4/4: I think that the answer to Mr. Perryman's question is that it is not a change in conductivity of the oxide which is affecting temperature, it is a change in the gas gap conductivity, the fuel can interface. The first effect is a deterioration in conductivity because of fission gas release. This cures itself as the fuel swells and gains contact with the can, so eventually the fuel can interface conductivity is improved and you get lower temperatures towards the end with a reduction in the columnar grain zone diameter.

E. C. W. PERRYMAN: At the beginning of the irradiation, if the diametral clearance is closing up, this will improve the heat transfer coefficient and I would expect the diameter of the columnar grain zone to decrease, not increase.

K. Q. BAGLEY: We are taking the opposite view. The first effect is release of fission products, xenon into the helium, which has a big effect on the gas gap conductivity, and causes the gradient across the gas gap to increase.

D. A. MOSS: I agree with Mr. Perryman on this. We did short term experiments on three capsules, varying from 2, 6, and 10 mils. These were a matter of hours and we find the gap has almost closed on the 10 mil. That is the radial gap. On the 6 and 2 mil, they closed almost immediately. The linear rating was 20 kW/ft.

H. STEHLE: Has anyone any results on creep of uranium oxide and mixed oxides and carbides, under irradiation and out of pile? Because I think the creep properties of the fuel would be important in regard to the direction in which they came. If the fuel creep is high, the can may withstand the swelling of the fuel. If the creep strength is higher for the fuel, the fuel will try to expand the can. Are creep experiments under way and, if so, are there any results?

H. LAWTON: We have not made any measurements in-pile of the creep behaviour of any of our ceramic fuels. You will remember, however, that I drew attention to the diminution in the hole size in pin A.153 which shows this creep under a combination of stresses occasioned by the fuel swelling outwards and the can holding it. I believe that this behaviour would be surprising if you were to do sums based on out-of-pile data. I cannot answer with regard to direct measurements or in-pile measurements of creep of these ceramics.

B. R. T. FROST: We have no direct evidence at Harwell although we have indirect evidence from the behaviour of UO_2 particles in cermets where, when they are highly restrained, they will flow into the voidage at temperatures as low as 600°C, whereas they obviously would not do that out-of-pile. We recognized that it is useful to know whether there is an irradiation effect on creep and have, in fact, designed two creep rigs, one for carbide and one for oxide. The carbide one is undergoing out-of-pile testing at present.

A. KARE HANNERZ (Sweden): If you have creep and a high coolant pressure, supercritical steam perhaps, in a weak cladding you can have compaction of the fuel, and if you have an annular pellet to start with, you can have shrinkage of the diameter and elimination of the centre of the void. This would perhaps not be acceptable. You can rely on coolant pressure to counteract the swelling in late fuel life. You can then get along with thinner cladding and high pressure coolant, but you have to stand up to depressurization, etc. I would like to raise this question and ask whether there are any comments on it.

K. KUMMERER: This question goes directly into the heart of the present thinking concerning the pin design for steam cooling. One of the major difficulties with pin design is the creep buckling problem of the cladding material and our standpoint at present is as follows: we do not want to depend on some mechanical properties of the fuel itself for having the right form of the fuel cross-section. We depend only on the strong can; also on a less strong can, and on steam cooling and the high pressures outside the steam. That means that for one solution we need very high strength cladding materials at high temperature, for example 750°C, and if you evaluate the different materials available, you will see that almost none of them can meet the requirements because in any case there is the problem of thermal stresses if you increase the thickness of the wall. Therefore, we propose to meet the requirements, to have an internal pressure of about 100 atm because we do not want to depend on the stiffness of the fuel. Then the outside pressure of the steam is about 150 or 170 atm. The answer at the moment would be that we see no chance to abandon the strong can concept in the steam cooled case.

K. KUMMERER: Dr. Bishop asked how we can get irradiation experience with such high canning temperatures. At present we are preparing for such experiments in the reactor at Karlsruhe where we are about to propose such a high temperature steam loop. He also asked if I regarded 750°C as acceptable for dynamic steam corrosion of Inconel 625. As far as in-pile corrosion is concerned, we have no experience. As far as out-of-pile corrosion is concerned, our experience has not reached such a high temperature. Therefore, in both cases, we still have to back up the hypothesis of our concept with experimental results.

O. SCHMIDT (Germany): I would like to comment on Dr. Kummerer's statement. First, there are some experiments going on with the superheat reactor which is planned. Secondly, it is 760° on the inside of the cladding at the hot spot so we are some 80° less at the outside.

K. KUMMERER: I have a question which deals with another fuel. There is some agreement that the right smeared density in the oxide fuel would be in the region of 80%, whereas we state 85%.

Different groups propose to have a fuel with this smeared density in the form of annular pellets. My view is that the main reason to take such an annular fuel is to avoid the high central temperature region.

Nobody needs such a high rod power to expect central melting because most of the reference designs are in the region of 500–600 W/cm, and with this fuel I think there is no danger of too much central melting. Therefore, it is not necessary to void the central region of the fuel.

On the other hand, if one has the same smeared density in the fuel, that means 8% for example, with an annular pellet, it means you have to intensify the fuel density up to 90% or so of the theoretical. Therefore, I would be afraid that, during irradiation, the swelling in the outward direction with such high density fuel would be more dangerous and more severe than with a fuel with the same smeared density, but distributed through all the cross-section.

The ceramic people and the fuel production people do not like to make such small pellets with 1 mm holes because they are less economical than normal pellets. I would ask the different groups or anyone else concerned with this philosophy for their comments.

J. F. W. BISHOP: The answer to the first question is, we would not agree that there is no concern in high temperature in the centre of fuel. We believe that this is a problem which will only get sorted out with much more irradiation experience. It is not necessarily melting one is concerned about, it is all sorts of other effects.

As regards your theoretical analysis of annular and vibro fuel, we would agree that the theoretical advantage of vibro fuel is the smeared density voidage towards the outside. You get worse gas retention, so you have an adverse effect on gas pressure but the improvement of lower density towards the outside and non-retained fission gas. In the annular case you do not get so much gas release as far as internal pressure is concerned but, in theory, you could well expect worse swelling in an identical model. Again we feel that the theoretical thinking on this has not gone far enough yet.

D. A. MOSS: I do not think we are completely clear how best to go about this. I would point out that whereas Dr. Kummerer's problem is making high density pellets, our problem seems to be making low density pellets. We

developed a technology to make very high density, 95–97%, and giving it controlled porosity of 80–85% gives us a quality control problem.

We have capsules under irradiation, both pellet and vibrating compacted fuel and are irradiating them under identical conditions. All these experiment are still in the reactor so we have a concern, too, I think.

We agree, possibly, that you can enhance the non-swelling properties of a fuel by removing the fission gases. This is why I do not know whether he will want to go to an annular fuel pin. The economics are not good. It might turn out to be a safety matter to have all the gas released, possibly during the normal situation, because it does not contribute very much to the fuel swelling problem.